Constraint-First Physics
a falsifiable route to the theory of everything

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Abstract

String theory's central obstacle is not that it predicts too little — it is that it appears to predict *nothing* testable, because the landscape of $\sim 10^{500}$ vacua cannot be searched or selected. This note states the result that reframes that obstacle as the answer: **the theory of everything predicts nothing exactly, except what it forbids (constraints) and what it is handed (labels)**. For the flux sector this is now a *proven theorem* — the only exact structure is the D3 tadpole, and the 20-year "search for the special vacuum" is provably dead. The frontier is the Standard Model: its exact structure is also constraint + label, and the open question — *are the 19 free parameters unpredicable in any ultraviolet completion?* — is a concrete, falsifiable target that a partner can pick up and settle either way. Everything quoted here is machine-verified.

1 The obstacle, restated

Two programs have tried to make the string landscape navigable. The *search* program asks whether we can learn the map from flux to observables and scan our way to our universe. The *constraint* program (swampland, bootstrap) asks which consistency conditions eliminate vacua. The difference between them is one measurable question: does any observable carry a *fingerprint* — strong information about the flux — that a search could exploit?

Measured on 174 real supersymmetric flux vacua, the answer is *no*. Every low-energy observable (string coupling, gauge couplings, mass-spectrum shape, axions, dilaton, cosmological constant) is statistically uniform in flux; exactly one quantity is exactly structured — the D3 tadpole, the constraint itself. The fingerprint the search program needs is absent.

2 What we proved: the theorem tower

The constraint-first claim began as that measurement and is now a tower of five theorems, each verified symbolically at zero tolerance, plus a machine-checked Lean arm (`pacland-proof/`). One command reproduces the whole tower (`scripts/validate_all_theorems.py`).

theorem	statement	grade
Mitchell's Uniqueness	the tadpole $N = F^\top J H$ is the *unique* bilinear symplectic invariant	6/6
Constraint-First	the tadpole generates the *entire* invariant ring (Weyl 1939, k=2)	5/5
Mitchell's Completeness	N is a *complete* invariant: same tadpole $\Leftrightarrow$ dual-equivalent	6+2
Full-duality	holds for $\mathrm{Sp}(b_3) \times \mathrm{SL}(2,\mathbb{Z})$, not just Sp	5/5
Complete Exact Structure	the ToE's exact content = constraints + labels, nothing else	assembly

The headline consequence: **the only exact selector in the flux landscape is the tadpole, and it is bounded.** A landscape search can only *eliminate* (by the constraint $N \leq L$); it can never *select*. The 20-year search program is provably dead — not because it failed, but because symplectic geometry forbids a fingerprint from existing. The claim is falsifiable and sharp: *a single moduli-independent, duality-invariant flux quantity that is not a function of $F^\top JH$ breaks the whole tower.* Nobody has to take it on faith.

3 The reframing: the ToE's exact content

Theorem 1 (Complete Exact Structure)**.** *Every exact (moduli-independent, duality-invariant) observable in a flux compactification is either a* ***constraint*** *(the tadpole, quantization, anomalies) or a* ***label*** *(the topology $h^{11}, h^{21}, c_2, \kappa$). Nothing else.*

In plain terms: the theory predicts nothing exactly except what it *forbids* and what it is *handed*. The quantities a theory of everything is conventionally expected to predict — the couplings, the mass ratios, the cosmological constant — are exactly the things that are *not* exact, and hence not predictable. That is not a bug in the framework; it is the answer.

4 The frontier: the Standard Model

The theorem's open question was *are all exact structures constraints?* The Standard Model is where that bites, and the answer so far is: *yes, with the same two-type decomposition.*

type	flux (proven)	Standard Model
constraint	tadpole $N \leq L$	anomaly cancellation, charge quantization
label	$h^{11}, h^{21}, c_2, \kappa$	gauge group $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$; $N_{\mathrm{gen}} = 3$
parameter	W_0, g_s, couplings, Λ	the **19 free couplings**

Two exact results are already established (both machine-verified):

1. **Anomaly cancellation is a constraint, not a prediction.** All five SM gauge/gravitational anomaly coefficients vanish exactly (`validate_sm_anomaly.py`, 5/5). The hypercharges (1/6, 2/3, −1/3, −1/2, −1) are *forced* by consistency, not chosen.
2. **The hypercharge is unique.** Given the SM matter content, the anomaly conditions determine the hypercharges uniquely up to overall normalization and the $u \leftrightarrow d$ swap (`validate_sm_hypercharge_uniqueness.py`, 4/4). This is the gauge-sector analog of Mitchell's Uniqueness — the *constraint* is unique.

5 The open target — and why it is worth picking up

The 19 free parameters (3 gauge, 9 masses, 4 CKM, 2 Higgs, 1 θ_{QCD}) are, within the Standard Model, free inputs *by definition* — that is a trivial counting fact, not a result. The non-trivial claim is:

Conjecture. No ultraviolet completion can predict the 19 parameters either — they are unpredicable in any theory of everything, for the same reason the flux couplings are: there is no invariant for them to be a function of.

This is the genuinely open, genuinely falsifiable frontier. The flux theorem was provable only because flux has a clean duality group (Sp) with a known invariant theory (Weyl's First Fundamental Theorem). The Standard Model's couplings have no such clean group action, so the conjecture is exactly that — and it may be *false*: a GUT or string construction that relates the couplings (gauge-coupling unification, a Yukawa relation) would be a counterexample.

Why a partner should care. This is a program with a success criterion, not a slogan. Either direction is a publishable, decisive result:

- *Confirmation:* exhibit the invariant-theory structure (the "duality group + FFT" analog for the gauge sector) that proves no invariant can fix the 19 parameters — completing the theorem in a second sector.
- *Falsification:* exhibit one ultraviolet relation among the 19 — which would itself be the kind of concrete prediction string theory has been accused of lacking.

Either outcome moves unification forward. The tooling is in place and open (`github.com/robmitchellengineering`); the four arXiv papers are ready; the verification discipline (a probably-approximately-correct certificate, the Mitchell Verification Deficit, the Rule of Three) is the same instrument used to keep the flux result honest, and it applies unchanged.

6 What is proven vs. what is open

claim	status
flux: the only exact structure is the constraint	**proven** (5 theorems + Lean)
SM: the constraint (anomaly → hypercharge) is unique	**proven** (4/4 exact)
SM: the 19 parameters are free inputs	**counted** (trivial)
SM: the 19 parameters are unpredicable in any ToE	**open, falsifiable**

The honest status of the program: the gravity sector's exact structure is a theorem; the gauge sector's exact structure is a theorem; the one sentence left — *"in every sector, the theory predicts nothing except constraints and labels"* — is proven in one sector and a falsifiable conjecture in the other. Closing it is the next theorem, and it has a clear yes-or-no test.